

FINANCIAL ANALYSIS

Introduction

The purpose of this project is to provide fundamental insights into how financial functions and terminologies work in Excel. It is essential to thoroughly understand the provided documentation and terms to efficiently utilize Excel's financial functions such as PV, NPV, XNPV, IRR, MIRR, and XIRR. This project involves analyzing a prepared dataset and applying these financial functions to derive meaningful financial insights.

Scope & Objectives

The supplied guideline defines the project as an Excel financial-analysis exercise covering present value, net present value, internal rate of return, effective interest rate, loan term and EMI, together with a summary of insights and analysis.

Methodology

Inputs and scenarios were taken from the supplied workbook and interpreted using the terminology in the guideline. Negative cash flows represent payments/outflows and positive values represent receipts/inflows. Formula-driven Excel calculations were built for PV, NPV, XNPV, PMT, IPMT, PPMT, CUMIPMT, CUMPRINC, RATE, NPER, IRR, XIRR and MIRR. Irregular cash-flow dates were ordered chronologically, and loan balances were reconciled.

Key Financial Concepts and Functions

Annuity - An annuity is a series of constant cash payments made over a continuous period, such as retirement savings, insurance payments, or mortgage payments.

In annuity functions:

- A positive number represents cash received.
- A negative number represents cash paid out.

Present Value (PV) - The present value is the total amount that a series of future payments is worth now.

Net Present Value (NPV) - NPV calculates the net present value of an investment based on a discount rate and a series of future payments and incomes.

XNPV - XNPV calculates the net present value for a schedule of cash flows that are not necessarily periodic.

Equated Monthly Installment (EMI) - An EMI is a fixed payment amount made by a borrower to a lender at a specified date each calendar month.

Interest and Principal Components of EMI - To calculate the interest and principal parts of the EMI:

Calculating Interest Rate - To find the interest rate required to pay back a loan.

Calculating Loan Term - To determine the number of payments required to clear a loan.

Internal Rate of Return (IRR) - IRR is the rate of interest at which NPV is zero.

XIRR - To calculate the IRR for irregularly spaced cash flows:

Modified IRR (MIRR) - MIRR takes into account different finance and reinvestment rates.

Methodology

Present Value (PV) - $PV = PV(\text{rate}, \text{nper}, \text{pmt}, [\text{fv}], [\text{type}])$

- rate: Interest rate per period.
- nper: Total number of payment periods.
- pmt: Payment made each period.
- fv: Future value (optional).
- type: Payment type (0 for end of period, 1 for beginning of period).

Net Present Value (NPV) - $NPV = NPV(\text{rate}, \text{value1}, [\text{value2}], \dots)$

- rate: Discount rate.
- value1, value2, ...: Cash flows.

XNPV - $XNPV = XNPV(\text{rate}, \text{values}, \text{dates})$

- rate: Discount rate.
- values: Cash flows.
- dates: Corresponding dates of the cash flows.

Equated Monthly Installment (EMI) - $PMT = PMT(\text{rate}, \text{nper}, \text{pv}, [\text{fv}], [\text{type}])$

- rate: Interest rate per period.
- nper: Total number of payment periods.
- pv: Present value or loan amount.
- fv: Future value (optional).
- type: Payment type (0 for end of period, 1 for beginning of period).

Interest and Principal Components of EMI –

- Interest Component: $IPMT = IPMT(\text{rate}, \text{per}, \text{nper}, \text{pv}, [\text{fv}], [\text{type}])$
- Principal Component: $PPMT = PPMT(\text{rate}, \text{per}, \text{nper}, \text{pv}, [\text{fv}], [\text{type}])$

Calculating Interest Rate - $RATE = RATE(\text{nper}, \text{pmt}, \text{pv}, [\text{fv}], [\text{type}], [\text{guess}])$

Calculating Loan Term - $NPV = NPER(\text{rate}, \text{pmt}, \text{pv}, [\text{fv}], [\text{type}])$

Internal Rate of Return (IRR) - $IRR = IRR(\text{values}, [\text{guess}])$

- values: Cash flows.
- guess: Initial guess (optional).

XIRR - $XIRR = XIRR(\text{values}, \text{dates}, [\text{guess}])$

Modified IRR (MIRR) - $MIRR = MIRR(\text{values}, \text{finance_rate}, \text{reinvest_rate})$

Implementations with Examples

Annuity

Suppose we are buying a refrigerator. The salesperson tells us that the price of the refrigerator is 32000, but we have an option to pay out the amount in 8 years with an interest rate of 13% per annum and yearly payments of 6000. We also have an option to make the payments either at the beginning or end of each year.

We want to know which of these options is beneficial for us.

Solution – We will use this Excel function for PV –

PV (rate, nper, pmt, [fv], [type])

- To calculate present value with payments at the end of each year, omit type or specify 0 for type.
- To calculate present value with payments at the end of each year, specify 1 for type.

If we make the payment now, we need to pay 32,000 of present value.

If we opt for yearly payments with payment at the end of the year, we need to pay 28, 793 of present value.

If we opt for yearly payments with payment at the end of the year, we need to pay 32,536 of present value.

We can clearly see that **option 2 is beneficial for us.**

Loan Repayment

Suppose, you want to take a home loan of 5000000 with an annual interest rate of 11.5% and the term of the loan for 25 years.

Rate per annum	0.12
Rate per month	0.01
Term	25
No. of monthly payments	300
Loan Amount (PV)	-5000000
FV	0
Type	1
EMI	₹ 52,139.81

We can find our EMI, here we will use PMT function to calculate EMI

As we observe,

- Present Value (PV) is the loan amount.
- Future Value (FV) is 0 as at the end of the term the loan amount should be 0.
- Type is 1 as the EMIs are paid at the beginning of each month.

We will get the following results –

- Monthly Payment of Principal and Interest on a Loan
- EMI includes both-interest and a part payment of principal.

As the time increases, these two components of EMI will vary, reducing the balance.

To get the interest part of your monthly payments, we can use the Excel IPMT function and the payment of principal part of your monthly payments, we can use the Excel PPMT function.

Rate per annum	0.16
Rate per month	0.013333333
Term	8
Loan Amount (PV)	-100000
FV	0
Type	0
EMI	₹ 13,261.59

Like, if we have taken a loan of 1,000,000 for a term of 8 months at the rate of 16% per annum. We can get values for the EMI, the decreasing interest amounts, the increasing payment of principal amounts and the diminishing loan balance over the 8 months. At the end of 8 months, loan balance will be 0. We can compute the interest and principal paid between two periods, inclusive by using the CUMIPMT function.

Calculating Interest Rate

Suppose, we take a loan of 100,000 and you want to pay back in 15 months with a maximum monthly payment of 12000. We might want to know the interest rate at which you have to pay.

Here we will use the Excel RATE function, where we will get the result as 8%

Calculating Term of Loan

Suppose, you take a loan of 100,000 at the interest rate 10%. You want a maximum monthly payment of 15,000. We might want to know how long it will take for you to clear the loan.

Here we will use the Excel NPER function, where we will get the result as 12 months

Decisions on Investments

When we want to make an investment, we compare the different options and choose the one that yields better returns. Net present value is useful in comparing cash flows over a period of time and deciding which one is better. The cash flows can occur at regular, periodical intervals or at irregular intervals. First, we consider the case of regular, periodical cash flows. The net present value of a sequence of cash flows received at different points in time.

At face value, Investment 1 looks better than Investment 2. However, we can decide on which investment is better only when you know the true worth of the investment as of today. We can use the NPV function to calculate the returns.

The cash flows can occur

- At the end of every year.
- At the beginning of every year.
- In the middle of every year.

NPV function assumes that the cash flows are at the end of the year.

We will get the following results –

Interest Rate	0.2	
	Cash Flow	
	Investement 1	Investement 2
Time		
1	-10000	-5000
2	25000	20000
3	-7000	-8000
Total	8000	7000

Timing	Investment 1	Investment 2
NPV (End Year)	₹ 4,976.85	₹ 5,092.59
NPV (Beginning Year)	5972.22	₹ 6,111.11
NPV (Middle Year)	₹ 5,451.87	₹ 5,578.66

As we observe NPV for Investment 2 is higher than that for Investment 1. Hence, Investment 2 is a better choice. We got this result as cash out flows for Investment 2 are at later periods as compared to that of Investment 1 Cash Flows at the Beginning of the Year

When the cash flows occur at the beginning of every year. In such a case, we should not include the first cash flow in NPV calculation as it already represents the current value. We need to add the first cash flow to the NPV obtained from rest of the cash flows to get the net present value.

When the cash flows occur in the middle of every year. In such a case, we need to multiply the NPV obtained from the cash flows by using sqrt function to get the net present value.

Cash Flows at Irregular Intervals

If we want to calculate the net present value with irregular cash flows, i.e. cash flows occurring at random times, the calculation is a bit complex. However, in Excel, we can easily do such a calculation with XNPV function.

Suppose today's date is 15th March, 2015. As we observe, all the dates of cash flows are of later dates. If we want to find the net present value as of today, include it in the data at the top and specify 0 for the cash flow.

Internal Rate of Return (IRR)

Internal Rate of Return (IRR) of an investment is the rate of interest at which NPV is 0. It is the rate value for which the present values of the positive cash flows exactly compensate the negative ones. When the discount rate is the IRR, the investment is perfectly indifferent, i.e. the investor is neither gaining nor losing money.

Cash Flows	Guess	IRR
10000		10.53%
-5000	0.05	10.53%
-8500	0.15	10.53%
2000	0.2	10.53%

As we can observe between the values of interest rate 10% and 11%, the sign of NPV changes. When you fine-tune the interest rate to 10.53%, NPV is nearly 0. Hence, IRR is 10.53. We can calculate IRR of cash flows with Excel function IRR. The IRR is 10.53% as you had seen in the previous section.

For the given cash flows, IRR may –

- exist and unique
- exist and multiple

If IRR exists and is unique, it can be used to choose the best investment among several possibilities. If the first cash flow is negative, it means the investor has the money and wants to invest. Then, the higher the IRR the better, since it represents the interest rate the investor is receiving. If the first cash flow is positive, it means the investor needs money and is looking for a loan, the lower the IRR the better since it represents the interest rate the investor is paying. To find if an IRR is unique or not, vary the guess value and calculate IRR. If IRR remains constant then it is unique.

As we observe, the IRR has a unique value for the different guess values.

In certain cases, you may have multiple IRRs.

Consider the following cash flows.

Cash Flows	Guess	IRR
-20000		-9.59%
82000	15.00%	-9.59%
-60000	20.00%	-9.59%
2000	25.00%	-9.59%
	30.00%	-9.59%
	35.00%	-9.59%
	40.00%	-9.59%
	45.00%	216.09%
	50.00%	216.09%
	55.00%	216.09%
	60.00%	216.09%

You can observe that there are two IRRs i.e., -9.59% and 216.09%. You can verify these two IRRs calculating NPV. For both -9.59% and 216.09%, NPV is 0.

Cash Flows	Guess	IRR
10000		#NUM!
-5000	0.05	#NUM!
8500	0.15	#NUM!
2000	0.2	#NUM!

In certain cases, you may not have IRR.

Consider the following cash flows. The result #NUM means that there is no IRR for the cash flows considered. The result #NUM means that there is no IRR for the cash flows considered.

Cash Flows Patterns and IRR

If there is only one sign change in the cash flows, such as from negative to positive or positive to negative, then a unique IRR is guaranteed. For example, in capital investments, the first cash flow will be negative, while the rest of the cash flows will be positive. In such cases, unique IRR exists. If there is more than one sign change in the cash flows, IRR may not exist. Even if it exists, it may not be unique.

Decisions based on IRRs

Many analysts prefer to use IRR and it is a popular profitability measure because, as a percentage, it is easy to understand and easy to compare to the required return. However, there are certain problems while making decisions with IRR. If we rank with IRRs and make decisions based on these ranks, we may end up with wrong decisions.

We have already seen that NPV will enable you to make financial decisions. However, IRR and NPV will not always lead to the same decision when projects are mutually exclusive.

Mutually exclusive projects are those for which the selection of one project precludes the acceptance of another. When projects that are being compared are mutually exclusive, a ranking conflict may arise between NPV and IRR. If we have to choose between project A and project B, NPV may suggest acceptance of project A whereas IRR may suggest project B. This type of conflict between NPV and IRR may arise because of one of the following reasons –

- The projects are of greatly different sizes
- The timing of the cash flows are different

Projects of significant size difference

Year	Project A	Project B
0	-1000	-1000
1	0	400
2	200	400
3	300	300
4	500	300
5	900	200
IRR	17.32%	20.49%
NPV	₹ 815.89	₹ 552.40

If you want to make a decision by IRR, project A yields a return of 100 and Project B a return of 50. Hence, investment on project A looks profitable. However, this is a wrong decision because of the difference in the scale of projects. If we invest entire 1000 on project A, we get a return of 100. If we invest 100 on project B, we will still have 900 in our hand that you can invest on another project, say project C. Suppose we get a return of 20% on project C, then the total return on project B and project C is 230, which is way ahead in profitability. Thus, NPV is a better way for decision making in such cases. Projects with different cash flows timings Again, if we consider IRR to decide, project B would be the choice. However, project A has a higher NPV and is an ideal choice.

IRR of Irregularly Spaced Cash Flows (XIRR)

Our cash flows may sometimes be irregularly spaced. In such a case, we cannot use IRR as IRR requires equally spaced time intervals. We can use XIRR instead, which takes into account the dates of the cash flow along with the cash flows.

Date	Cash Flow
08-04-2015	-10000
15-08-2015	4000
15-03-2016	3000
25-04-2016	5000

The Internal Rate of Return that results in is 26.42%

Modified IRR (MIRR) - Consider a case when your finance rate is different from your reinvestment rate. If you calculate Internal Rate of Return with IRR, it assumes same rate for both finance and reinvestment. Further, you might also get multiple IRRs. For example, consider the cash flows given below where finance rate is 10% and reinvestment rate is 12%–

Year	Cash Flow
0	-1.6
1	10
2	-10

Discount	NPV
10%	-\$0.70
25%	\$0.00
110%	\$0.43
400%	\$0.00
500%	-\$0.04

As you observe, NPV is 0 more than once, resulting in multiple IRRs. Further, reinvestment rate is not taken into account. In such cases, we can use modified IRR (MIRR), we will get a result of 7%.

Key Findings

Annuity and Present Value (PV): At 13% with eight annual payments of Rs 6,000, the end-of-year payment stream has PV - Rs 28,792.62; the beginning-of-year stream has PV – Rs 32,535.66. The beginning-of-year PV is higher because payments occur earlier. The end-year PV is below the Rs 32,000 cash price in the supplied scenario.

Loan and EMI: The 25-year Rs 50 lakh loan at 12% has an EMI of approximately Rs 52,139.81. The eight-month Rs 1 lakh loan at 1.3% monthly has an EMI of approximately Rs 13,242.27. The amortization schedule demonstrates the expected decline in interest and increases in principal repayment over time. The RATE example implies a monthly rate of 8.44%, while the NPER example gives approximately 11.53 months.

Investment NPV: At a 20% discount rate, Investment 1 NPV is Rs 5,972.22 and Investment 2 NPV is Rs 6,111.11; Investment 2 is preferred. The model also distinguishes end-, beginning- and middle-of-period timing. The guideline emphasizes NPV where mutually exclusive projects differ in size or cash-flow timing.

XNPV: The irregular-date cash-flow schedule produces an XNPV of approximately Rs 17,523.65 at 20% when dates are evaluated chronologically. XNPV is appropriate because the cash flows do not occur at equal intervals.

IRR & XIRR: The standard IRR example produces approximately 10.53%. The supplied guideline also demonstrates that multiple sign changes can generate multiple IRRs, including approximately -9.59%

and 216.09% in its example. XIRR is the appropriate extension for irregular dates. Consequently, IRR should be interpreted alongside NPV and checked for multiple roots.

MIRR: With a 10% finance rate and 12% reinvestment rate, MIRR is approximately 6.55%. MIRR separates the finance and reinvestment assumptions and is useful when conventional IRR is ambiguous.

Actionable Recommendations

- Use NPV as the primary decision metric for mutually exclusive investments when scale/timing differs.
- Use PMT/IPMT/PPMT for loan affordability and repayment analysis.
- Use XNPV/XIRR for irregularly dated cash flows.
- Investigate multiple IRRs before relying on IRR rankings.
- Keep assumptions, timing and sign conventions documented in financial models.
- Prefer formula-driven models so assumptions can be changed without rebuilding outputs.

Conclusion

We can conclude that:

- Timing changes present value.
- Loan EMIs combine interest and principal, with the interest share declining over time.
- Investment 2 has the higher NPV in the supplied 20% scenario. • XNPV/XIRR are appropriate for irregular dates.
- Multiple IRRs can arise from multiple cash-flow sign changes.
- MIRR provides a controlled return measure with separate rates.